BMS Logics for Operation

1. Charge/Discharge MOS Control Description

By default, the main charging and discharging MOSFETs are in the off-state. They are controlled via three specific pins:

- ACC (System Wake & Paralleling): Applying P+ to the ACC pin wakes up the system and provides a limited output to power the Vehicle Control Unit (VCU). When two batteries are connected, triggering ACC also enables their parallel operation logic.

- DIN (Discharge Control):
  Applying P+ to DIN turns on the main discharge MOSFET(and turns off the charge MOSFET). This must be activated whenever the system's power requirement exceeds the 2A provided by ACC. ACC and Din both are needed to be short together to perform large discharge.

- CIN (Charge Control): Applying P+ to CIN turns on the charging MOSFET. For the BMS to actually accept charging current, two strict conditions must be met:The CIN MOSFET must be ON, and a valid battery charging CAN communication signal must be received. (When CIN and Din are both shorted at the same time, the priority goes to Charging, however, when Cin is active, discharge is limited to 2A supply power to VCU).

2. Deep hibernation and shallow hibernation mode

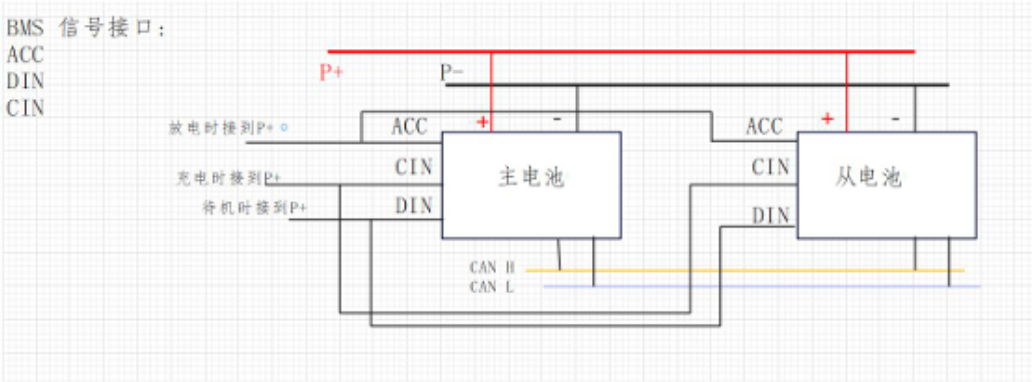
- Deep Hibernation

Our system enter deep hibernation only when the battery is deep discharged (cell voltage reaches to shutdown voltage of around 2.75V)
Deep hibernation Wake up: Automatic activation upon power-on; activation requires CIN connection to P+ in deep sleep mode.

- Shallow Hibernation

No current , no equalization, communication interruption , 10-second delay before entering shallow sleep
Shallow sleep wake-up conditions: current , communication , CIN , DIN , ACC (any one of these conditions must be met).

3. Parallel and switching logic



Use the serial number (SN) for comparison.

Determine the capacity level in the SN; the larger capacity level indicates the master, and the smaller capacity level indicates the slave.

If the capacity levels are the same, determine the production date and serial number; the one with the smaller number is the main unit.

Master-slave : Master is 0x14, slave is 0x1D.

The serial number (SN) is only compared when both ACC and battery are connected simultaneously. If one battery already has a serial number of 0x14/1#, the next battery connected will have a serial number of 2#0x1d. The comparison is repeated after disconnecting ACC and connecting both batteries simultaneously.

When a single group is connected via ACC, it sends a serial number (SN). If it does not receive the SN of another battery, it automatically becomes battery #1/0x14

Discharge Strategies:

When there is no command control, the main battery should be discharged first when ACC is connected.
If the main battery level drops to below 5%, switch to discharging the secondary battery. This will cause the secondary battery to discharge till 0%. Then again switch to utilize the final 5% from the main battery. This is done as the main battery is the master which controls the switching to secondary. The secondary can never give a command switch.

If, during the switch, the battery group to be switched to is in an undervoltage (or discharge protection) state while the discharging battery is not undervoltage, a switch failure will be reported, and the switch will not be performed; continue placing the current battery

If, after the command to switch is executed, the current battery level drops to <5%, the system will automatically switch to another group, and the command to discharge the current battery will become invalid.
If the battery group to be switched to is in an undervoltage (or discharge protection) state, do not switch and continue discharging the current battery. If the battery being switched to is not in a state of discharge protection, the switch is successful and the battery is discharged to the point of over-discharge.

Charging Strategy:
When the charger is connected, it will prioritize charging the battery with the lowest voltage until it is fully charged, and then switch to the other battery group to fully charge it.